



AROR UNIVERSITY
OF ART, ARCHITECTURE,
DESIGN & HERITAGE,
SUKKUR, SINDH

Department of Cyber Security, Artificial Intelligence, Multimedia and Gaming

Course Title: Applied Physics (Theory)

Program & Class: Cyber Security-II	Semester: (II)
Credit Hours: 03 (02+01)	Instructor: Engr- Danish Asad
Office location: Faculty Offices Building-1, Room-204	Post-requisite Courses: None
Office hours: 9 AM – 5 PM Consulting hours: Tuesday - (2 PM – 3 PM) Thursday - (11 AM – 1 PM)	E-mail. danish.faculty@aror.edu.pk

COURSE DESCRIPTION:

Motion along a straight line, Vectors, dot product and cross product. Motion in 2 and 3 dimensions. Force, Friction, Work and Energy, Kinetic and Potential energy, Conservation of energy, Waves, Vibrations and Oscillations, Simple Harmonic Motion, Wave Motion and Sound, Mechanics, Electric Charge, Conductors and Insulators, Coulomb's Law, Quantization and Conservation of Charge, Electric Fields, Introduction to Electric Field, A point charge in electric field, A dipole in electric field, Gauss' Law, Electric Flux, Gauss' Law and its Applications, Electric Potential, Electric potential and Electric potential energy, Potential due to a point charge, Potential due to group of charges, Potential due to an electric dipole, Potential due to continuous charge distribution, Capacitance, Introduction to capacitance, Capacitors in parallel and series, Energy stored in an electric field, effect of Dielectric on capacitance, Current and Resistance, Introduction to electric current, Effects of Electric Current. Sources of electricity, Current density, Resistance and Resistivity, Ohm's Law, Power in electric circuits, Semiconductors and super conductors, Introduction to electric circuits, Pumping charges, Work, energy and EMF, Single and Multi-loop circuits, RC circuit, capacitive time constant, Semi-conductor Physics

COURSE LEARNING OUTCOMES:

CLO1: Illustrate the electromagnetic and mechanical phenomenon mathematically

CLO2: Interpret the basic electric circuits used in electrical engineering

CLO3: Apply the knowledge of Resistors, Capacitors and Inductors to design and implement basic circuits using modern tools and techniques.

CLO4: Apply the knowledge of electromagnetics and mechanics to perform experiments, analyze data and its interpretation.

CLO – PLO Mapping:

CLOs	Related PLOs	Taxanomy	CLO attainment checked in
CLO1	1	Cognitive Level 2	Quiz/Assignment, Mid Exam, and Final Exam
CLO2	1	Cognitive Level 3	Quiz/Assignment, Mid Exam, and Final Exam
CLO3	2	Psychomotor Level 3	Mid Lab Exam, and Final Lab Exam
CLO4	4	Psychomotor Level 3	Mid Lab Exam, and Final Lab Exam

ASSESSMENT/ EVALUATION (Theory)

1.	Mid Term Exam	30
2.	Sessionals	30
3.	Final Examination	40
Total		100

RECOMMENDED BOOKS:

S.No	Book Name	Author/s Name	Publisher Name & Edition
1.	Fundamentals of Physics	David Halliday, Robert Resnick, and Jearl Walker	Tenth Edition, John Wiley & Sons, ISBN: 0471465097.
2.	Electronic Devices (Conventional Current Version)	Thomas L. Floyd	10 th Edition

REFERENCE BOOK:

S.No	Book Name	Author/s Name	Publisher Name & Edition
1.	University Physics	Hugh D Young and Roger A. Freedman	15 th Edition

Important Policies:

The student is expected to attend all of the scheduled classes if for some reason the student cannot make a class, he/she must inform in advance via email only, The course includes Assignment, Quizzes, Midterm Exam and a Final Exam . The student is expected to turn in all work on time.

WEEK WISE DETAILS:

Week	Topics	Assignments/ Quizzes	Suggested Readings
01 - 02	Vectors, Force and Motion: Position and Displacement, Average Velocity and Average Speed, Acceleration, free fall acceleration, scalars and vectors, addition, components of vectors, unit vectors, multiplying vectors, projectile motion, analysis of projectile motion, the horizontal and vertical motion, the equation of path, the horizontal range, effect of the air. Force, some particular forces, applying newton laws, friction, properties of friction, the drag force and terminal speed.		Book [1]
03 - 05	Work and energy: Work and Energy: Kinetic energy, work, Work and kinetic energy, Work and Potential Energy, Determining Potential Energy Values, Conservation of Mechanical Energy, the center of mass, The Rotational Variables, Are Angular Quantities Vectors? Kinetic Energy of Rotation, linear momentum, the linear		Book [1]
	Momentum of a System of Particles, Conservation of Linear Momentum, torque revisited, the angular momentum of systems of particle, conservation of angular momentum.	Quiz -01 (7.5 Marks)	Book [1]
06 - 07	Waves: Simple Harmonic Motion, The Force Law for Simple Harmonic Motion, Simple pendulum and physical pendulum, types of waves, Transverse and Longitudinal Waves, The Wave Equation, The Principle of Superposition for Waves, Interference of Waves.		Book [1]
08	Electric Charge: Coulomb's law, Conductors and Insulators, conductor with a cavity, charge quantization, charge		Book [1]

	conservation.		
09 - 10	Electric field: introduction to electric field, Field due to a point charge: electric field of Dipole. Electric field due to line of charges. Gauss's law: Electric flux Gauss's law and its application. Electric Potential: Electric potential and Electric potential energy Potential due to a point charge, Potential due to group of charges, Solve numerical problems involving topics covered	Quiz – 02 (7.5 Marks)	Book [1]
11 - 13	Capacitance: Introduction to capacitance Capacitors in parallel and series, Energy stored in an electric field, Dielectric. Current and Resistance: Introduction to electric current, Effects of Electric Current, Sources of Electricity, Current density, Resistance and Resistivity, Ohm's Law, Power in electric circuits, Semiconductors and super conductors. Circuits: Introduction to electric circuits, Pumping charges, Work, energy and EMF, Single and Multi-loop circuits, The ammeter and voltmeter capacitance, dielectrics. Growth and Decay of voltage in an RC circuit. Solve numerical problems involving topics covered.	Quiz – 03 (5 Marks)	Book [1]
14 - 16	Electronics: Basic crystal structure, free electron model, energy band in solid and energy gaps, doping, p-type, n-type semiconductor materials, p-n junction diode, its structure. Characteristics curve of a diode, half wave rectification, full wave rectification, peak inverse voltage in case of full wave rectifier, characteristics and application as rectifiers. Transistor, its basic structure and operation. Solve problems on NPN Transistor	Assignment (Hand Written) (10 Marks)	Book [2]